

Automated Financial Risk Assessment: NLP Methods for Severity Classification and Risk Factor Identification

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1 Problem statement and motivation

Financial risk assessment is a fundamental challenge in modern finance, which requires organizations to analyze large volumes of corporate reports, SEC filings, and earnings disclosures to identify potential financial risks. Traditionally, this process has relied on manual expert review, which is time-intensive, inconsistent, and prone to human bias. As the markets become more complex, the need for automated, scalable, and accurate risk assessment tools has grown. The US Securities and Exchange Commission (SEC) alone receives over 700,000 disclosure filings annually, highlighting the increasing volume of data that must be analyzed ([U.S. Securities and Exchange Commission, 2022](#)).

Advancements in Natural Language Processing (NLP) have introduced new possibilities for automating financial risk classification, severity prediction, and sentiment analysis in financial documents. Although sentiment analysis in financial texts has received substantial research attention, risk severity classification remains comparatively underexplored. This gap stems from several challenges, such as the limited availability of labeled risk severity datasets. Unlike sentiment data which can be derived from market reactions or analyst ratings, risk severity labels often need domain expert annotation, creating significant barriers to research.

In this project, we explore NLP techniques to demonstrate that machine learning models and neural networks can extract meaningful signals from financial texts, improving decision-making compared to traditional methods. The aim is to apply NLP techniques to automatically classify risk severity level reported in corporate disclosures and to explore methods that may help identify textual drivers behind the risk severity level and address the technical gap.

2 Research hypothesis

Our research hypothesis assumes that domain-specialized transformer-based NLP models, FinBERT, will significantly outperform traditional machine learning approaches and general-purpose transformer language models (for example RoBERTa) in classifying financial risk severity. This hypothesis is based on recent advancements in transformer architectures that have demonstrated superior performance in capturing contextual relationships in text compared to machine learning approaches. Financial text contains complex semantic relationships and specific terminology that transformer models are particularly adept at processing. Furthermore, key phrase extraction and interpretation techniques will provide meaningful insights into model decision-making.

Confirming this hypothesis would demonstrate the importance of domain-specific pretraining for specialized financial text analysis tasks in automating financial risk assessment task. We will evaluate this hypothesis using a comprehensive dataset of corporate disclosures with annotated risk levels and compare a range of approaches including traditional machine learning methods (Logistic Regression and Support Vector Machines with Bag-of-Words, TF-IDF and Word2Vec as feature vectorizers) and transformer-based models (RoBERTa and FinBERT). These methods represent a spectrum from simple statistical approaches to advanced neural architectures, allowing us to comprehensively assess the relative advantages of domain specialization in financial risk classification tasks.

3 Related work and background

The application of natural language processing to financial documents represents a growing yet still underexplored area of research, particularly for specialized tasks like risk severity classification; largely due to the scarcity of publicly avail-

able annotated datasets in this domain. (Arslan et al., 2021) presented a comprehensive comparison of pre-trained language models for multi-class text classification in the financial domain. Their study evaluates generic models (BERT, DistilBERT, RoBERTa, XLNet, XLM) against a domain-specific model (FinBERT) on both benchmark and proprietary financial datasets. Their findings notably challenge the assumption that domain-specific models outperform generic ones, as FinBERT did not demonstrate superior performance even on financial texts. However, the author did not apply any methods to optimise performance of FinBERT. A generic transformer like RoBERTa could serve as a strong baseline for financial text classification, while highlighting the importance of proper fine-tuning rather than assuming domain-specific models will automatically perform better.

(Araci, 2019) demonstrated that transformer-based language models pre-trained on financial text significantly outperform traditional approaches for domain-specific classification tasks, which aligns with our hypothesis that FinBERT should outperform the other models. From their analysis, their FinBERT model achieved 15% accuracy improvement over state-of-the-art methods on financial sentiment analysis while requiring substantially less training examples. Their research showed that fine-tuning only the last few transformer layers yield comparable performance with combined training strategies (like gradual unfreezing, discriminative learning rates). These findings support using pre-trained language models for risk severity classification with interpretability methods like SHAP.

(Wang et al., 2024) presented a comprehensive NLP-based approach which systematically details text preprocessing, feature extraction (BoW, TF-IDF, word embeddings), and model construction techniques applied to financial documents. Their empirical results demonstrate that LSTM models (92.5% accuracy) outperform traditional approaches for identifying potential financial risks. Their preprocessing pipeline and evaluation framework provide valuable methodological insights. Their model comparison strategy and metrics (accuracy, precision, recall, F1) offer a solid foundation for evaluating my own augmentation techniques. This paper confirms the effectiveness of NLP techniques in specialized domains where textual data analysis is critical.

(Shorten et al., 2021) showed how text augmentation techniques can be particularly valuable in scenarios where data is limited. Their analysis of Easy Data Augmentation (EDA) revealed that synonym replacement offers substantial performance gains, especially with limited data. They found more improvements with just 500 samples and minimal gains when samples surpass 5000 samples. They used symbolic methods like rule-based and neural augmentations like back-translation and provided empirical evidence of their effectiveness. For imbalanced datasets, they argued that augmentation could be a better alternative to traditional sampling methods. Their findings suggest that synonym replacement could effectively address class imbalance by strategically expanding minority classes while avoiding duplication.

(Masson and Paroubek, 2024) applied unsupervised topic modelling on annual reports by French financial institutions. Using LDA, NMF and Top2Vec, they found that traditional methods produced incoherent topics. Top2Vec showed the best performance, capturing semantically meaningful topics. However, according to them, interpretability was challenging. Their work highlights limitations of unsupervised methods in financial NLP. In contrast, my project uses supervised severity classification and explainability tools which would generate interpretable insights from labelled risk disclosures.

(Yin et al., 2023) introduced a novel approach, FinPT, that transforms financial data into natural language profiles using LLMs, then fine-tunes for risk prediction. The authors also develop FinBench, a comprehensive benchmark for financial risk assessment. Experiments demonstrate FinPT's superior performance over traditional methods across default, fraud, and churn prediction tasks. Particularly impressive is the Flan-T5-Base implementation, which achieves the highest F1-scores on most datasets. The approach bridges tabular data analysis and natural language processing, leveraging the semantic understanding capabilities of large language models. This work addresses significant gaps in financial risk prediction by modernizing methodologies and establishing standardized evaluation metrics.

(Theil et al., 2018) presented a significant contribution to financial uncertainty detection by automatically expanding the Loughran-McDonald dictionary using word embeddings. They compared

general-domain and domain-specific embedding models, revealing that domain-specific expansions significantly improve the detection of uncertain sentences in 10-K reports which in turn better explain stock return volatility. The authors find that manual expert filtering provides minimal additional benefit over fully automated approaches. These findings support the use of domain-specific language models and automated lexicon expansion for risk classification tasks in financial documents.

(Martínez et al., 2025) advanced the field of financial NLP by addressing a critical gap in regulatory text understanding. Their work is motivated by the increasing complexity of regulatory documentation and the inadequacy of general-purpose LLMs for domain-specific applications in finance and regulatory compliance. The authors' primary contribution is a comprehensive framework for creating domain-specific language models that balance computational efficiency with performance. They employed a strategic approach to data collection and preprocessing, combining recursive web scraping of official regulatory sources with a TF-IDF relevance filtering pipeline to create a high-quality corpus. This methodological innovation helps overcome the challenge of noisy web-sourced data while maintaining the specialized vocabulary and structures found in regulatory texts.

(Govindaraj et al., 2023) presented an innovative approach integrating transformer models with explainability mechanisms for financial forecasting. Their architecture combines SHAP values and attention visualizations to enhance interpretability without significantly sacrificing predictive accuracy. The study demonstrates how transformer attention layers can effectively capture temporal dependencies in financial data while making predictions explainable through feature importance rankings and attention maps. This research is particularly relevant for my risk severity classification project, as their SHAP implementation could directly enhance interpretability of FinBERT predictions.

(Tan and Kok, 2024) represents a significant contribution to the ever increasing field of explainable artificial intelligence in finance. They address the critical limitation of current financial risk assessment models which is lack of comprehensive explanation for risk classification. They introduced FinBERT-XRC, a novel explainable deep learning model to address the limitation of current models. Their model combined FinBERT's contextual lan-

guage understanding with a hierarchical attention mechanism to classify company risk based on 10-K reports. The model's primary innovation is its three-level explanation framework, which provides interpretability at word, sentence, and corpus levels simultaneously. Their model outperformed state-of-the-art XRR model. Their approach to explainability is particularly valuable for financial applications where understanding risk factors is crucial. The model's success with financial text also reinforces the value of domain-specific pre-trained language models like FinBERT for capturing the specialized vocabulary and contextual relationships in financial documents.

3.1 Accomplishments

- Task 1: Initial analysis of the dataset to determine proper preprocessing steps - Completed
- Task 2: Preprocess dataset (removing noise and converting to lowercase) - Completed
- Task 3: Tokenise the dataset and apply Lemmatization – Completed
- Task 4: Apply text vectorization (BoW, TF-IDF) for base models – Completed
- Task 5: Build and train base models and examine performance results with different text vectorization methods – Completed
- Task 6: Optimize performance of base models and examine performance – Completed
- Task 7: Explore pre-trained (GloVe) word embeddings - Completed
- Task 8: Build and train BERT models – Completed
- Task 9: Conduct experimental fine-tuning to optimize BERT models – Completed
- Task 10: Experiment with Key Phrase extraction methods - Completed
- Task 11: Apply optimized base model and best BERT model on Test Data for evaluation – Completed
- Task 12: Perform in-depth error analysis to identify examples that possibly led to misclassifications – Completed

- Task 13: Apply SHAP/LIME to identify key factors – Completed
- Task 14: Apply selected key phrase extractor on Test data to analyze important phrases without relying on the models – Completed

4 Approach and Methodology

In this project, we adopt a methodology that combines both traditional and deep-learning approaches to financial text classification. For our baseline models, we employ Bag of Words (BoW) and Term Frequency-Inverse Document Frequency (TF-IDF) for interpretable feature extraction which is widely supported by prior research in financial text analysis. To enrich semantic understanding, we also incorporate Global Vectors for Word Representation (GloVe) embeddings.

BoW is a simple text representation method that converts a document into a vector based on frequency or presence of each word, ignoring grammar and word order. Given that this approach ignores word order, it cannot capture the meaning of words or the relationship between words. However, it is easy to implement and highly interpretable.

TF-IDF is another text vectorization method that weighs words based on how important they are to a document relative to a collection of documents. TF-IDF highlights rare, meaningful terms that carry important financial signal (Wang et al., 2024). However, TF-IDF would still ignore word order which may limit its ability to understand semantic meaning in text.

GloVe is a word embedding method for turning words into dense vectors where words that appear in similar contexts across a large corpus are placed closer together in a vector space. It captures semantic similarity. In finance, this helps to better capture latent financial topics and risks that simpler frequency-based methods (BoW/TF-IDF) might miss (Govindaraj et al., 2023). However, GloVe embeddings are static, words could have different meanings depending on the context but would be assigned only one vector. Word2Vec was considered but we ultimately opted for GloVe for its ability to incorporate global corpus statistics which would be better for capturing the nuanced and systemic relationships present in financial documents.

The extracted text features are then used to train traditional machine learning models: **Support Vector Machine (SVM)** and **Logistic Regression**.

SVM is noted for excellent classification performance, especially for small sample size and high-dimensional data (Wang et al., 2024). Logistic regression is a traditional but very effective for structures feature spaces like TF-IDF. We also considered Naïve Bayes, but we have seen that it struggles as it assumes feature independence which may not always hold for financial documents.

For our deep-learning approaches, we explore **FinBERT** and **RoBERTa**. FinBERT is a domain-specific language model based on **BERT** architecture and is fine-tuned specifically for financial texts. FinBERT was selected due to its strong ability to capture contextualized financial semantics, making it particularly well-suited for risk prediction and sentiment classification tasks within the financial domain. Text inputs were processed using the FinBERT tokenizer, a WordPiece-based tokenizer, which splits text into subword units optimized for financial vocabulary. The use of FinBERT allowed for a deeper semantic understanding of complex financial disclosures compared to traditional static embeddings like GloVe. We employ RoBERTa which is a general-purpose deep language model to contrast the performance of a domain-agnostic model against a domain specific model like FinBERT. Text inputs were tokenized using Byte-Pair Encoding (BPE) tokenizer designed for RoBERTa, which efficiently handles a wide vocabulary without being specialized for financial terminology.

Depending on models used, we implement **Explainable AI (xAI)** to enhance the interpretability of the best-performing machine learning and deep learning models. For the best traditional machine learning model, we implement **LIME** which generates locally linear approximations of complex models which helps us understand which features contribute strongly to each prediction. For deep learning models, we use **SHAP**, a game theoretic approach. SHAP provides a global and local view of feature impact, making it well-suited to interpret the nuanced decisions made by complex transformers. By applying LIME and SHAP, we aimed to balance high predictive performance with transparency, offering insight into both the traditional and deep learning models' decision-making processes.

We also experimented with **KeyBERT**, an unsupervised keyword and key phrase extraction method. KeyBERT serves as a tool to extract the most important financial terms without relying on

any model training. The idea is to compare the SHAP/LIME (what the model thinks is important) against KeyBERT (what the document emphasizes) to validate whether the models are learning the right financial risks.

Throughout the project, we utilize a combination of standard and specialized Python libraries to support data processing, modelling, and explainability. Core libraries include NumPy and Pandas for data manipulation, and Matplotlib. For traditional machine learning models, we employ Scikit-learn. Deep learning components, including FinBERT and RoBERTa implementations, were built using PyTorch and the HuggingFace Transformers library. We also used NLTK as part of processing the data.

5 Dataset

The dataset of choice is the Gretel Financial Risk Analysis v1, which consists of a collection of financial analysis texts, generated synthetically and is available on (AI, 2024). It was trained on 14,306 SEC filings (10-K, 10-Q, and 8-K) from 2023 to 2024, aiming to facilitate the extraction of key risk factors and the generation of structured summaries from financial documents. The total samples consist of 1034.

5.1 Dataset preprocessing

Two key findings from the initial data exploration directly influence the preprocessing strategies employed.

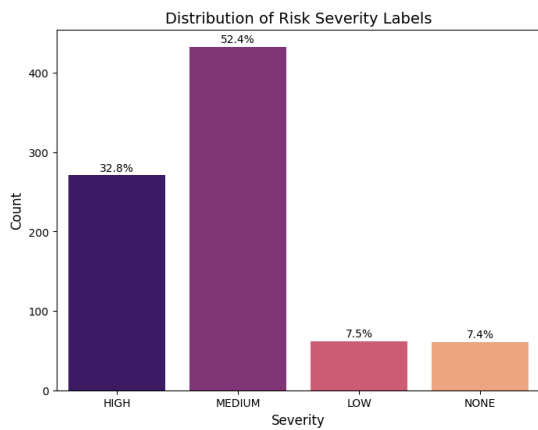


Figure 1: Risk severity distribution across the dataset.

The class distribution plot reveals significant class imbalance, with ‘HIGH’ and ‘MEDIUM’ risk levels dominating the dataset. To address this imbalance, we apply data augmentation strategy. More specifically, we apply synonym replacement

on the minority classes (‘LOW’ and ‘NONE’) to synthetically increase their representation. The augmentation process involves randomly selecting words within minority class samples and replacing a subset of them (approximately 20%) with synonyms using the WordNet lexical database. This technique help introduce lexical diversity while preserving the original meaning of the texts (Team, 2022).

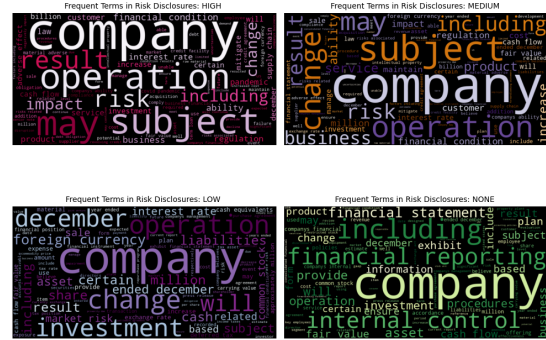


Figure 2: WordClouds of risk severity levels.

Through WordCloud analysis, we observe a set of extremely common domain-specific words (e.g. “company”, “report”) that frequently appears across risk levels. These words carry limited value for risk prediction. Consequently, an extended stop word list is manually curated to remove these domain-specific but non-informative terms during text preprocessing, refining the feature space for downstream modelling.

Beyond data augmentation and stop word refinement, we also apply text normalization where all text is converted to lowercase along with financial metric standardization to minimize noise. Texts were tokenized using ‘word_tokenize’ operation from NLTK to split documents into individual words or tokens. We also make sure to preserve key financial terms such as ‘dollar’, ‘percent’, and ‘date’. A lemmatization step, using the ‘lemmatizer’ operation from WordNet, is applied to reduce words to their base forms, reducing the dimensionality of features and improving the generalization of similar terms.

6 Baselines

To establish baseline performance, multiple model combinations have been evaluated. Specifically, Bag-of-Words (BoW), Term Frequency-Inverse Document Frequency (TF-IDF) and GloVe are used text vectorization methods, each combined with

either Logistic Regression (LR) or Support Vector Machine (SVM). The models are evaluated based on accuracy and F1-score, given the imbalance in the dataset. We observed overfitting due to small imbalanced training data. Among these, the TF-IDF with SVM model achieves the best performance. As a result, this model is selected for further hyperparameter tuning using Grid Search. However, the tuned model shows only marginal improvements compared to the untuned baseline, indicating that the initial model was already relatively well-optimised for the task. The results from GloVe are unexpectedly very low, probably due to potential mismatch between the embeddings and the domain-specific language used in financial disclosures.

Table 1: Base Models performance (Accuracy)

	BoW	TF-IDF	GloVe
Logistic Regression	0.71	0.69	0.56
SVM	0.72	0.75	0.47

Pre-trained Transformer models, specifically RoBERTa and FinBERT, are fine-tuned on the financial risk severity dataset. Initial experiments revealed slight overfitting, indicated by a gap between training and validation performance. To address this, regularization techniques such as dropout, weight decay and early stopping are applied to improve generalization. These methods helped stabilize training and achieve better performance on unseen data. During training, the FinBERT model achieved accuracy 0.73 after fine tuning compared to 0.70. For RoBERTa, we initially observed 0.69 accuracy and 0.71 accuracy after fine tuning.

7 Results, error analysis

7.1 Baseline Model (TF-IDF + SVM)

After evaluating several classical approaches, the TF-IDF with SVM is selected as the baseline. The model achieved an achieved an accuracy score of 0.75 and F1-score of 0.77 during training. When the model is evaluated on the test set, we observe that class imbalance issue persists. The model performs well on the ‘MEDIUM’ class, achieving an F1-score of 0.72 and reasonably well on the ‘HIGH’ class with F1-score of 0.67. The model struggles significantly with the minority classes despite text augmentation.

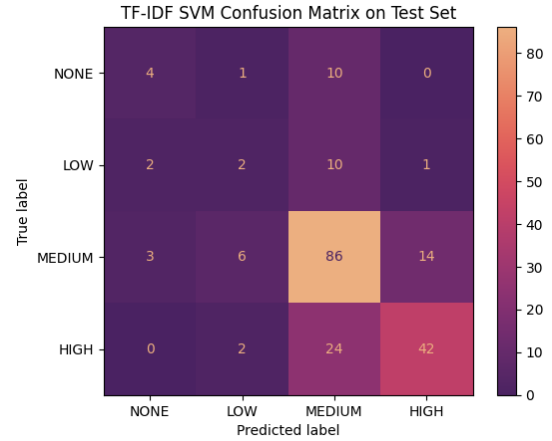


Figure 3: TF-IDF+SVM Confusion Matrix.

Looking at the confusion matrix, we can further confirm the issue of class imbalance. The model shows bias towards the majority classes, correctly classifying 86 out of 109 samples for the ‘MEDIUM’ class and 42 out of 68 of the ‘HIGH’ class. The model has not learned enough about the minority classes due to insufficient samples. The model is defaulting to ‘MEDIUM’ when uncertain, hurting the performance on minority classes.

7.1.1 Error Analysis

- **HIGH Severity Document misclassified as LOW**

We examined of the cases where the model incorrectly classified high-risk severity as low severity. The case we observe is sample 199 from the dataset which describes SEC investigation settlement with \$150M in penalties. It includes disgorgement of ‘ill-gotten gains’ and civil money penalty. There are also requirements for enhancements to internal controls over financial reporting and review of their accounting and disclosure practices. The document also contains significant regulatory consequences to certify completeness of its disclosures. These are all indicators of a high-risk disclosure. Despite the clear markers of high severity, the model potentially predicted low severity level as in the document we observe a formal disclosure language that downplays the severity through phrases such as:

- “will not have a material impact on the Company’s liquidity”
- “does not require the Company to admit or deny allegations”

- “allows the Company to focus on its core business”

This pattern suggests the model may be overly influenced by mitigating language rather than recognizing the substantive severity indicators.

• LOW Severity Document misclassified as MEDIUM

To analyze why the model predicted medium severity for a low-risk document, we look at sample 157. In the document, it is mentioned that routine statements about risk management practices are performed. We have tabular data about the financial position of the company which shows normal values. There is also mention of financial metrics such as “\$1.5 billion of cash and cash equivalents” and “\$800 million of short-term investments”. There is the presence of neutral statements about interest rate exposure. These are no indicators of financial distress. However, the model misclassified the document due to financial terms that would usually be found in high-severity disclosures such as “impairment” or “deferred tax assets and liabilities”. Large dollar amounts could also have triggered higher severity classification. We also noticed risk-related vocabulary such as “affected by changes in interest rates”.

7.1.2 LIME

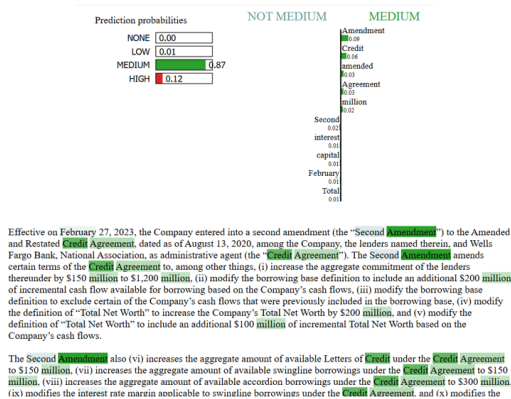


Figure 4: TF-IDF+SVM LIME Analysis.

LIME analysis shows that the model heavily weighs legal terminology (‘Amendment’, ‘Credit Agreement’) and financial amounts (‘million’) when assessing severity, rather than the actual impact of these changes. This explains why routine credit facility amendments are often misclassified as MEDIUM severity despite representing normal

business operations. The visualization reveals the model assigns high importance to formal financial terminology (0.09 for “Amendment”, 0.06 for “Credit”) while overlooking contextual signals that would indicate low severity. This pattern suggests the model has developed a bias toward certain financial and legal terms commonly found in financial disclosures regardless of their context.

7.2 BERT Model (FinBERT)

After fine-tuning RoBERTa and FinBERT, we identified that FinBERT showed the best performance on our dataset. The domain-specific FinBERT achieved an overall accuracy of 0.66 and F1-score of 0.51 on the test set. The fine-tuning process adapted the pre-trained financial language model to our specific severity classification task.

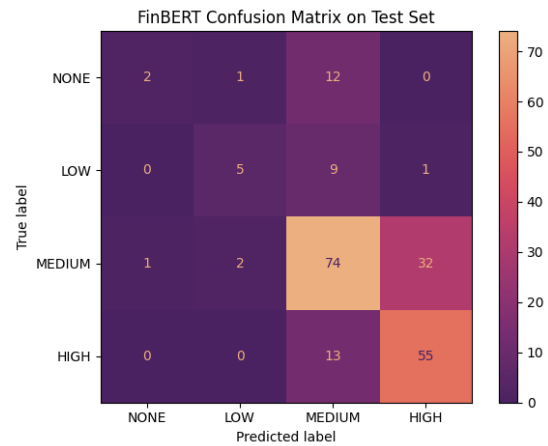


Figure 5: FinBERT Confusion Matrix.

Looking at the performance metrics, FinBERT demonstrated variable effectiveness across severity classes. The model performed strongest on high-severity disclosures (F1=0.71, recall=0.81), effectively identifying 55 out of 68 high-risk documents. The model also showed balanced performance for the medium-severity class with an F1-score of 0.68.

However, despite fine-tuning, FinBERT continued to struggle with minority classes, achieving lower F1-scores for “NONE” (0.22) and “LOW” (0.43) severity documents. The confusion matrix reveals a consistent pattern of severity escalation, particularly misclassifying lower-severity documents as “MEDIUM.”

When compared to our TF-IDF with SVM approach (Accuracy: 75.3%, F1: 77.1%), the fine-tuned FinBERT model showed lower overall performance despite its specialized pre-training on financial texts. While FinBERT excelled at high-severity

document detection with superior recall, the SVM model demonstrated more balanced performance across all severity classes, particularly for minority classes but misclassified two high-severity documents as low-severity. This performance gap highlights that traditional machine learning approaches with careful feature engineering can sometimes outperform complex language models.

7.2.1 Error Analysis

- **NONE Severity Document misclassified as MEDIUM**

For error analysis, we look at the sample with index number 26 where true label is ‘NONE’ and the model predicted ‘MEDIUM’. Throughout the text, the document repeatedly states that various accounting standard adoptions ‘did not have a material impact on the Company’s consolidated financial statements.’ The document describes purely routine accounting policy updates required by new accounting standards (ASUs), with no indication of financial distress, regulatory issues, or material financial consequences. Those are no signs for high level financial risk.

FinBERT most likely misclassified the document due to the presence of technical accounting terminology such as ‘revenue recognition policies’ or ‘performance obligations’. Phrases such as "implemented new policies and procedures" and "updated its internal controls" might be associated with remediation of accounting problems in FinBERT’s pre-training data. Words like "changes," "amendments," and "impact" appear throughout the document which could impact the decision of the model.

- **HIGH Severity Document misclassified as MEDIUM**

We look at sample 140 to analyse what may have caused the misclassification as ‘MEDIUM’ rather than ‘HIGH’ severity. The document contains information about regulatory breach where market capitalization is mentioned to have fallen below \$40 million. They also received official notice from NYSE and are warned about delisting procedures. There is also mention that the company is working to address other challenges including decline in revenue and profitability. These are tell-tale signs of high-severity financial disclosure. Despite those indicators, there are various statements in the document that downplays the severity:

- “The Company intends to continue to work to address the NYSE’s concerns”
- “The Company’s management and board of directors are committed to maintaining the highest standards”
- “The Company is actively exploring options to improve its market capitalization”

The document also does not specify direct financial losses or immediate material impacts. The model struggles when documents use standardized disclosure language that intentionally presents information in a measured and non-alarming manner.

7.2.2 SHAP



Figure 6: FinBERT SHAP Analysis.

SHAP analysis revealed that classification decisions of FinBERT heavily rely on specific financial terminology, with terms like "amendment," "entered," and "administrative agent" receiving substantial weighting in the model’s decision process. The analysis provided quantitative measures of how these terms influence classification, with output scores demonstrating their relative importance.

However, applying SHAP to FinBERT faced a significant limitation due to the 512 token constraint of transformer models. This forced truncation of longer financial documents, potentially missing critical information in later sections that might influence classification. The example used demonstrates this limitation, as only the opening section could be analyzed, potentially missing key severity indicators appearing later in the document.

Compared to LIME, SHAP’s numerical approach offers more mathematical rigor but lacks LIME’s intuitive in-text highlighting that clearly identifies specific phrases influencing classification decisions.

7.3 KeyBERT

To complement the model-specific explainability techniques, we implement KeyBERT as a document-level semantic analysis tool that extracts keywords and key phrases without relying on model predictions. This approach is particularly valuable given SHAP’s 512-token limitation with

FinBERT and LIME’s focus on individual terms rather than semantic relationships.

For our sample document, phrases extracted from KeyBERT include: "agreement modifies" (0.74), "million incremental" (0.64), and "prudent responsible" (0.62). Comparing these results with LIME and SHAP revealed:

1. KeyBERT identified semantic relationships between terms (e.g., “agreement modifies”) that individual term analyses miss.
2. It captured business implication phrases (“fund operations,” “prudent responsible”) not explicitly highlighted by model-specific techniques.
3. Unlike SHAP, KeyBERT analysed entire documents without token limitations.
4. It provided semantic context to financial metrics that aligned with but expanded upon LIME’s term emphasis.

However, KeyBERT also extracted temporally specific but risk-irrelevant phrases (e.g., "august 13") and sometimes produced generic business terminology lacking direct risk implications.

This multi-method approach demonstrates that combining statistical significance (LIME), contextual financial understanding (SHAP), and semantic relationship extraction (KeyBERT) provides a more comprehensive view of financial risk factors than any single method alone.

8 Lessons learned and conclusions

The project on automating financial risk assessment using HLP techniques provided several key take-aways. We have seen that traditional methods like TF-IDF with SVM can still outperform domain specialised transformer models. The dataset, being relatively small with uneven distribution, led to biased model predictions. Techniques like data augmentation helped but were insufficient, underscoring the importance of collecting or synthesizing larger, balanced datasets to improve model generalization. We have also seen that financial documents very often utilise mitigating languages to undermine the severity of their position; this suggests that models need enhanced contextual understanding or specialised training to differentiate between tone and substantive risk indicators. Applying, explainable AI tools like SHAP or LIME are essential to

build trust in the automated systems; we have seen that models rely on formal financial terminology (e.g. “amendment”, “credit agreement”) in their predictions. KeyBERT to extract keyphrases independently of model predictions provided a useful benchmark for validating SHAP and LIME outputs. However, its unsupervised nature sometimes produced less relevant phrases, indicating a need for supervised or semi-supervised methods tailored to financial risk factors.

This area of text classification is underexplored due to the lack of annotated data. The results observed could serve as a benchmark for future research on the topic. Larger and balanced data would help the model distinguish exclusive patterns and semantic relationships among the different risk severity documents. For future implementation, I would suggest finding more data on minority classes through expert annotation or even the use of LLMs to generate synthetic ones. Traditional methods should still be explored along with transformer models as they have shown good performance throughout this project. Model interpretability is important to develop robust and trustworthy tools for automated financial risk assessment.

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